Mushroom Edibility Classification Using Machine Learning

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Abstract

Mushroom picking is a popular activity that carries significant health risks due to the difficulty of distinguishing between edible and poisonous species based on morphological similarities. The primary objective of this study is to develop a robust machine learning capable of correctly classifying the edibility of mushrooms using categorical features. A secondary but equally critical goal is to ensure the transparency and interpretability of the model's decision-making process. Using the "Mushroom Classification" dataset, we compared the performance of Decision Tree, Naive Bayes, and Random Forest classifiers. The Random Forest model achieved the best results with nearly perfect accuracy. To prioritize user safety, the classification threshold was calibrated to eliminate false negatives. In addition, Explainable AI (XAI) techniques, specifically SHAP (SHapley Additive exPlanations), were applied to identify key discriminating features. The results show that the smell (*odor*) and color of the spores (*spore-print-color*) are the most critical predictors.

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1 Introduction

It has happened to everyone, at least once, to encounter wild mushrooms and wonder whether they are safe to eat. However, without specific knowledge, distinguishing edible from poisonous species is extremely difficult.

The increasing adoption of machine learning models in scientific domains has led to a development of increasingly accurate models for classification. In particular, in the field of mycology, where the correct identification of edible and non-edible (poisonous) mushrooms represents a significant problem, both for public health protection and for supporting collection and research activities.

Traditionally, mushroom identification relies on morphological and sensory analysis that requires expert knowledge. However, these approaches are prone to error, especially in the presence of species with highly similar characteristics.

The practical importance of this problem is highlighted by the large number of mushroom poisoning incidents reported every year. Looking at the data collected and published by "Adnkronos Salute", in Italy alone, we have around 10000 cases of mushroom intoxication, 90% of which are registered during the autumn harvesting season. These figures underline the limitations of manual identification approaches when morphological similarities between species are high.

A representative example is Tylopilus felleus, commonly known as the bitter bolete or gall bolete, which closely resembles several edible porcini species (Boletus edulis group) in terms of size, cap color, and general structure.

Figure 1: Tylopilus felleus on the left and Boletus edulis on the right

Despite its harmless appearance, Tylopilus felleus is not edible, but the strong morphological similarity makes it particularly difficult for inexperienced collectors to distinguish it from its edible counterpart.

In this context, the use of supervised machine learning techniques provides an effective alternative, enabling the analysis of large datasets and the discovery of complex patterns that are difficult to detect using conventional methods. By framing mushroom identification as a supervised classification problem, we trained and evaluated different models capable of generating reliable predictions on edibility based on the relationship between mushroom features.

1.1 Research Questions

This study is structured around two fundamental research questions that guide our analytical approach:

1. **Model Development**: Can we develop a machine learning model capable of accurately classifying mushrooms as edible or poisonous based solely on categorical morphological features? What modeling approach achieves the best predictive performance in terms of accuracy, precision, recall, and robustness?

2. **Feature Interpretation**: Which morphological characteristics best distinguish between edible and poisonous mushrooms, and how do they influence the classification decision? Specifically, we aim to identify not only which features are most important, but also to understand the mechanism through which different values of these features contribute to the final prediction.

To address the first question, we employ a comparative analysis of multiple classification algorithms, evaluating their performance through rigorous cross-validation and test set evaluation. For the second question, we move beyond traditional statistical association measures (such as Cramer's V and contingency tables) to apply explainable AI techniques, specifically SHAP (SHapley Additive exPlanations), which provides a comprehensive, model-based understanding of feature importance and their directional effects on predictions.

2 Exploratory Data Analysis

A fundamental phase is exploratory data analysis, which aims to understand the main characteristics of the dataset, identify potential patterns, and formulate initial hypotheses that will subsequently be examined.

The dataset chosen for answering the research question is Mushroom Classification, available on the Kaggle platform. Although the health statistics cited refer to the Italian context, this dataset, which includes descriptions of 23 species taken from The Audubon Society Field Guide to North American Mushrooms (1981), was chosen for its completeness and validity as a standard for testing mycological classification algorithms. It originally contained 23 variables and 8124 observations representing different mushrooms and their morphological characteristics.

2.1 Preliminary Inspection

The analyzed dataset consists only of categorical variables. The preliminary inspection made it possible to analyze the data type of each variable, confirming that all variables are of type "String". This analysis is essential for the subsequent phases, particularly for the construction of machine learning models, which will require appropriate encoding and imputation procedures for textual variables.

2.2 Variance Analysis

The variance analysis revealed that the variable *veil-type* has a variance equal to zero, indicating that all observed values are identical. This characteristic makes the variable non-informative; therefore, it was considered redundant for the purposes of the study and removed during the preprocessing phase.

2.3 Correlation Analysis

To analyze the relationships among categorical variables, a correlation matrix based on the chi-square test was constructed. This approach makes it possible to assess the existence of statistically significant dependencies between variables, providing useful insights into the underlying structure of the dataset.

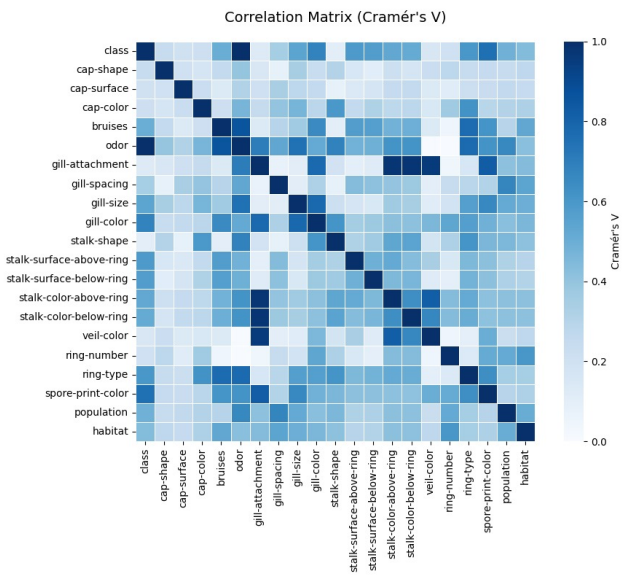

Figure 2: Chi-square based correlation matrix

As shown in Figure 2, the correlation matrix visualizes the strength of the relationships between variables. Dark blue shading corresponds to values close to 1, indicating a strong correlation, while lighter shades correspond to values close to 0, indicating no correlation. An important pattern emerges from the graph analysis: the target variable *class*, which indicates whether

a mushroom is edible or poisonous, exhibits a high correlation with the *odor* variable. This finding suggests that *odor* has high discriminative power for classification purposes. We expect the *odor* feature to be one of the most important predictive variables in machine learning models.

2.4 Target Variable Distribution

A final key analysis before proceeding with preprocessing is the examination of the target variable distribution, aimed at verifying the balance between classes. The following table reports the absolute and percentage frequencies of observations for each class.

Class	Frequency	Percentage
Edible	4,208	51.8%
Poisonous	3,916	48.2%
Total	8,124	100.0%

Table 1: Target variable distribution in terms of absolute and percentage frequencies

From the table, it emerges that the dataset consists of 3,916 poisonous mushrooms and 4,208 edible mushrooms, corresponding to approximately 48.2% and 51.8% of the total observations, respectively. This distribution indicates a reasonably balanced dataset, reducing the risk of biased learning toward one class. As a simple reference, a naive majority-class classifier would achieve an accuracy of approximately 51.8%, highlighting the need for non-trivial models to capture meaningful patterns.

3 Preprocessing

The dataset underwent a structured data preprocessing implemented primarily in KNIME, with a preliminary step performed in Python, where the string "?" was globally replaced with NaN values to standardize the representation of missing values. During this process we noticed the presence of the string "None" that didn't represent a missing but a class, like in the case of the *odor* and *ring-number* variables where it represented odorless and ringless mushrooms.

The preprocessing started with the encoding of the target variable *class* into a binary format (0-1) using the String Replacer node.

The variable *veil-type* was eliminated since it contained a single category constant across all observations thus adding nothing to the analysis.

An additional feature, *stalk-root*, was removed because it had a high proportion of missing values (more than 30%), specifically 2480 missing values out of 8124 observations. Imputing this variables was deemed inappropriate, as it could have significantly distorted the analysis results.

During the process two checks for duplicate observations were performed to identify any potential duplicates as to avoid data redundancy, however none were found.

4 Feature Importance

After the preprocessing phase, an exploratory analysis of feature importance was conducted to identify the categorical variables most strongly associated with the target variable. Since all the predictors are nominal, the association between each feature and the target was measured using Cramer's V, computed through the linear correlation node in KNIME.

This measure is based on the chi-square statistic and it quantifies the strength of association between two categorical variables, with values ranging from 0 (no association) to 1 (perfect association). Cramer's V provides a quantitative measure of the discriminative power of each categorical feature with respect to mushroom edibility. The analysis revealed that only a small subset of features shows a strong association with the target variable, most notably *odor*, *spore-print-color*, *gill-color*, and *ring-type* with corresponding values of 0.971, 0.753, 0.681 and 0.603.

4.1 Contingency Tables

To better understand the nature of these relationships, some contingency tables were constructed between each of these variables and the target. Of these tables, only the contingency table related to *odor* (Table 2) is reported in the main text, due to its near-deterministic separation power. The other tables are provided in Appendix A for completeness.

These tables highlight the presence of perfect or almost perfect separation for some feature categories, especially for the *odor* variable where smells like anise and almond always represent edible mushrooms while other categories like foul, spicy, fishy, pungent and musty correspond exclusively to poisonous mushrooms. This phenomenon indicates that certain categorical values are highly predictive of the target.

Class	almond	anise	creosote	fishy	foul	musty	odorless	pungent	spicy	Total
0	400	400	–	–	–	–	3,408	–	–	**4,208**
	9.51%	9.51%	–	–	–	–	80.99%	–	–	
1	–	–	192	576	2,160	36	120	256	576	**3,916**
	–	–	4.90%	14.71%	55.16%	0.92%	3.06%	6.54%	14.71%	
Total	400	400	192	576	2,160	36	3,528	256	576	**8,124**

Table 2: Cross Tabulation of Class by Odor

Such separation effects motivated the choice of avoiding logistic regression models as they would encounter convergence issues. Perfect separation violates the underlying assumptions of maximum likelihood estimation, leading to infinite or unstable coefficients estimates and preventing the optimization algorithm from converging. Consequently, models capable of handling categorical splits directly were preferred.

5 Models

To ensure a robust comparison between different classification approaches, a 5-fold cross-validation strategy was adopted. This procedure reduces the risk of overfitting and allows for a more reliable estimation of model performance by repeatedly training and testing the models on different subsets of the data. The use of cross-validation is particularly valuable given the relatively limited size of the dataset, as it reduces variance in performance estimates and ensures that every observation contributes to both training and evaluation.

5.1 Decision Tree Classifier

This model creates a hierarchical structure of binary splits based on feature values, creating a tree-like model of decisions. Each internal node represents a test on a feature, each branch represents the outcome of the test and each leaf node represents a class label. The classification rule is the path from root to leaf.

The selected split criterion was the Gini impurity index and the pruning method was Minimum Description Length (MDL) that searches for a balance between a deep tree model that better adapts to the training data but risks overfitting and a simple tree that is easier to understand but makes more error in the classification. A light condition on the minimum number of records per node was also applied to stop the split if one or both the nodes created by it had less than 20 records respectively.

When evaluated on the Test set, the decision Tree achieved good overall classification accuracy (mean accuracy: 0.997) demonstrating an ability in distinguishing meaningful patterns from the categorical features. Its high recall (mean recall: 0.997) is particularly relevant in our case as we want to minimize the false negatives that are poisonous mushrooms erroneously classified as edible.

5.2 Naive Bayes Classifier

Naive Bayes is a probabilistic classifier based on Bayes' theorem with the "naive" assumption that features are conditionally independent given the class label. For each observation, the model computes the posterior probability of each class and assigns the observation to the class with the highest probability. The independence assumption is rarely true in practice but the model often yields surprisingly good results due to the robustness of the classification decision even when probability estimates are inaccurate. Considering our datasets and the strong dependencies observed in the feature importance step this assumption could be problematic so we must look at the results with more attention.

The default probability, minimum standard deviation, and threshold standard deviation were all set to 0.0001, while the maximum number of unique nominal values per attribute was set to 20.

The model showed a slightly lower performance than the decision tree, both in accuracy and recall (mean accuracy: 0.975, mean recall: 0.974), that is likely attributed to the violations of the independence assumption. Considering mushrooms it is clear that some features have strong interdependencies because they are often biologically related, for example cap characteristics like color, shape and surface. These dependencies mean that the product of individual conditional probabilities doesn't accurately represent the joint probability distribution.

5.3 Random Forest Classifier

The Random Forest is an ensemble method that constructs multiple decision trees during training and outputs the class that is the mode of the classes predicted by individual trees. Each tree is trained on a random sample extracted from the original training set and at each split only a random subset of features is considered introducing additional noise. This combination of randomness reduces correlation between trees and improves generalization performance as it reduces overfitting. A disadvantage of this model is its difficult interpretation as the decision rule is a combination of different rules and as such it is often too complex for a human to grasp.

The split criterion selected for the training of the 100 tree models is the information gain criterion, the tree depth was set at 10 and the minimum node size was 20 again.

As expected random forest consistently outperformed both decision tree and naive bayes across all metrics, demonstrating superior predictive capability and stability. The model showed accuracy and recall approaching 1 (mean accuracy: 0.999, mean recall: 0.999) indicating a notable ability in correctly identifying poisonous mushrooms, a critical capability for this application. All the results of the evaluations were collected into the following table and a bar chart with a focused range between 0.9 and 1.0 in order to better highlight the differences among the mean metrics of the compared models.

Metric	Tree	Naive Bayes	Random Forest
Recall	0.997	0.974	0.999
Precision	0.997	0.976	0.999
Sensitivity	0.997	0.974	0.999
Specificity	0.997	0.974	0.999
F-measure	0.997	0.975	0.999
Accuracy	0.997	0.975	0.999
Cohen's Kappa	0.994	0.958	0.998

Table 3: Model performance comparison

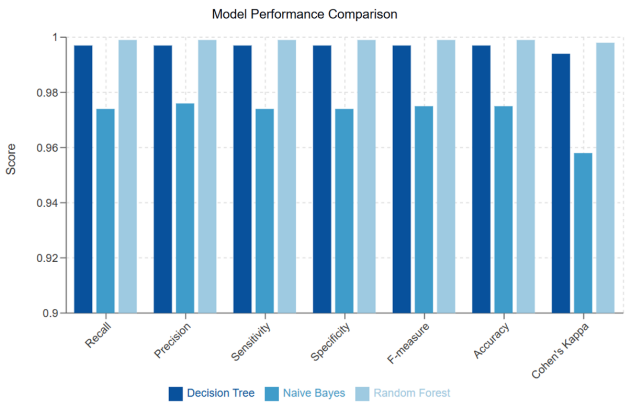

Figure 3: Model evaluation comparison

6 Tuning

The ROC curve was computed for the best-performing model, Random Forest, to evaluate its discriminative capability across different classification thresholds. The resulting ROC curve shows a steep ascent toward the upper-left corner, corresponding to an AUC (Area Under the Curve) of approximately 0.999, indicating nearly perfect discriminative capability between edible and poisonous mushrooms.

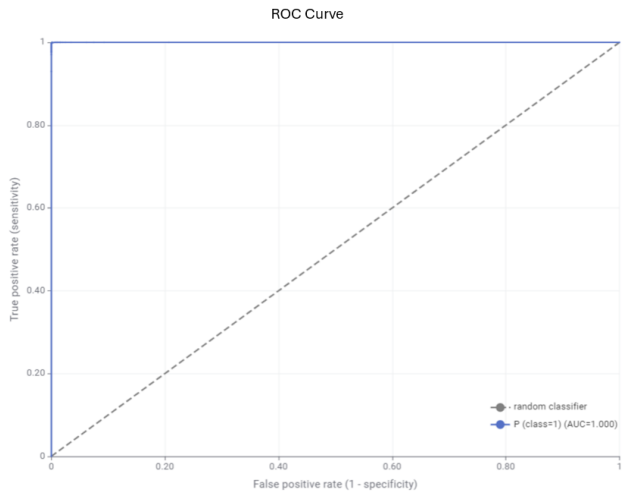

Figure 4: Roc Curve Graph

By examining both the ROC curve and the confusion matrices obtained at different thresholds, a critical decision was made regarding the classification threshold. The standard threshold of 0.5 yielded 8 false negatives (poisonous mushrooms misclassified as edible) and 0 false positives (edible mushrooms misclassified as poisonous). However, from a food safety perspective, false negatives represent the most dangerous type of error, as they could lead to the consumption of poisonous mushrooms with potentially severe health consequences.

For this reason, the classification threshold was deliberately lowered from 0.5 to 0.15. This adjustment resulted in a complete elimination of false negatives (from 8 to 0), while increasing false positives from 0 to 26. Although this trade-off might seem unfavorable from a purely statistical standpoint, it is entirely justified in the context of mushroom classification: misclassifying an edible mushroom as poisonous (false positive) leads only to a missed opportunity for consumption, whereas misclassifying a poisonous mushroom as edible (false negative) could result in serious intoxication or even death.

This threshold adjustment aligns the model's behavior with the primary objective of the research, prioritizing user safety by maximizing sensitivity (recall for the poisonous class) at the expense of specificity. The choice reflects a risk-averse approach that is appropriate for critical classification tasks where the cost of different types of errors is highly asymmetric.

7 Explainable AI

Explainable AI (XAI) is an approach that aims to make the decision-making process of black box models transparent and understandable. This method allows users to trust the results, verify their reliability, and comply with regulatory requirements. In summary, the basic concept of XAI is to clarify the behavior of the model by revealing which features drive its predictions and how.

One of our key research questions was to identify which morphological categories are most relevant for distinguishing between edible and poisonous mushrooms. While the contingency tables presented earlier provided initial insights into feature-target relationships, we sought a more comprehensive and model-based understanding of feature importance. For this reason, we applied XAI techniques to obtain a more complete assessment that accounts for feature interactions and their context-dependent contributions within the trained model.

One of the most well-known and robust approaches in the XAI field is SHAP (SHapley Additive exPlanations), which is grounded in cooperative game theory. Specifically, in our study, SHAP was applied to interpret the results of the best-performing model, identifying not only the most important morphological features but also revealing how different values of these features influence the probability of a mushroom being edible or poisonous.

The analysis of global feature importance reveals which variables are the most influential in the model's final decision, thereby helping to understand which morphological characteristics are most critical for classification. The beeswarm plots summarize how each feature contributes to the model predictions, displaying both the direction (toward edible or poisonous) and the magnitude of the effect on class probability. It is important to note that the features on the y-axis are arranged in descending order of global importance based on mean absolute SHAP values; therefore, the variables at the top have the strongest overall impact on the model's output.

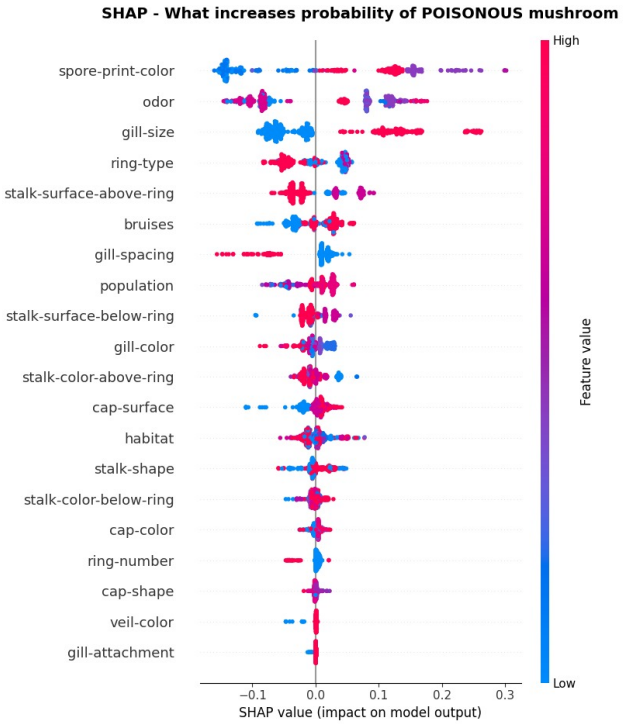

Figure 5: SHAP beeswarm plot for the poisonous class

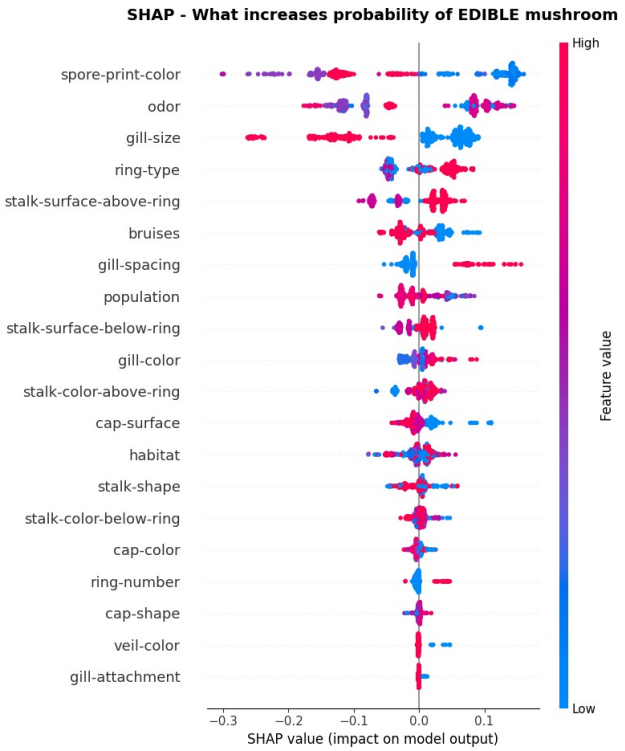

Figure 6: SHAP beeswarm plot for the edible class

- Class 'POISONOUS': From Figure 5, the beeswarm plot displays each feature's impact on predicting poisonous mushrooms. The x-axis represents the SHAP value, which indicates the magnitude and direction of each feature's contribution to the model's output: positive values (right side) increase the probability of the mushroom being poisonous, while negative values (left side) decrease it. Each point represents an individual observation, with the color encoding the feature value according to the legend (high values in red/pink, low values in blue). For instance, high values of *spore-print-color*, *odor*, and *gill-size* (red points) are predominantly located on the right side of the plot, indicating that these specific feature values strongly increase the probability of a mushroom being classified as poisonous. The vertical dispersion of points at each SHAP value level shows the density of observations with similar contributions.

- Class 'EDIBLE': Figure 6 reveals an opposite pattern compared to the previous class. Here, positive SHAP values (right side) indicate an increased probability of the mushroom being edible. Notably, high values (red points) of *spore-print-color*, *odor*, and *gill-size* are predominantly positioned on the left side, corresponding to negative SHAP values. This means that these same high feature values that contributed positively to the poisonous class now contribute negatively to the edible class, which is consistent since the two classes are complementary. Conversely, low values (blue points) of these features appear on the right, increasing the probability of edibility.

Based on the Cramer's V analysis conducted during the exploratory phase, we expected the most relevant features for the model to be *odor*, *spore-print-color*, *gill-color*, and *ring-type*. The SHAP analysis yielded results that are largely consistent with these expectations, though with some notable differences. In particular, *gill-size* appears among the top features in place of *gill-color*, despite the latter showing a higher Cramer's V value (0.681). This substitution can be attributed to the high correlation between these two variables, as evidenced by the correlation matrix presented earlier. Since SHAP values measure the marginal contribution of each feature to the model's predictions, *gill-size* likely provides a slightly higher marginal contribution than *gill-color* when both are considered together. This highlights a fundamental difference in how these two methods evaluate feature importance: while Cramer's V measures the overall strength of association between a feature and the target variable independently, SHAP values assess the incremental impact of each feature given the presence of all other features in the model. Consequently, when two features are highly correlated, SHAP tends to assign greater importance to the one that offers superior additional predictive power in the context of the complete feature set.

8 Final discussion and conclusions

The main goal of this study is to determine whether it is possible to classify a mushroom as edible or not, using exclusively morphological characteristics. To achieve this goal, different machine learning algorithms were used to confirm this hypothesis. The Random Forest model is the most effective approach, reaching accuracy and recall values of nearly 100%. An interesting and important result has been achieved by XAI and EDA. In both cases, the feature *odor* and *spore-print-color* have been identified as the most important features in the discrimination of the target class.

8.1 Limits of the study

Despite the excellent performance achieved by the machine learning models, it is fundamental to analyze the results while considering some limitations that affect the model's applicability and generalizability.

The most important limitation is that the dataset dates back to 1981 and refers specifically to North American mushroom species. Therefore, the model's geographical generalization capability is limited. For this reason, using this model to classify European or Asian mushrooms may not be as accurate and may lead to misclassifications.

Furthermore, the model's classification accuracy is fundamentally dependent on the user's ability to correctly identify and describe the morphological characteristics of the mushroom in question. This introduces a critical human factor in the prediction pipeline: even a highly accurate model will produce incorrect results if fed with inaccurate or imprecise feature values.

For these reasons, while the model demonstrates strong predictive performance within its intended scope, it should not be used as a standalone tool for mushroom identification in real-world foraging scenarios. Instead, it could serve as a complementary decision support system to be used alongside traditional mycological expertise and field guides.

8.2 Future developments

The understanding of the limits of the study allows the identification of alternatives to improve analyzes. The first solution concerns the limitations of the sample. The data may be integrated with a more modern dataset and a more even geographical distribution. This will help make the models created more robust on a global level. Another future development concerns the integration of computer vision techniques to reduce subjectivity in the input of data. Currently, the accuracy depends on the user's ability to describe the mushroom correctly. The use of Convolutional Neural Networks (CNN), allows for the automatic extraction of morphological characteristic directly from the images, minimizing human error while keeping a certain level of user control. This approach may lay the foundation for the development of applications that provide a preliminary assessment in real time.

8.3 Conclusions

In conclusion, this study shows that machine learning techniques, particularly Random Forest combined with Explainable AI, can effectively support mushroom edibility classification through accurate and interpretable models. However, the results must be interpreted within the limits of a curated dataset of North American species with clear morphological separations, which may not fully capture the complexity of real-world scenarios. Furthermore, model performance depends on correct human encoding of features, which is not guaranteed in practice. Despite these limitations, the proposed system represents a promising decision-support tool that, with further data and validation, could contribute to reducing the risk of mushroom poisoning.

A Additional Contingency Tables

Class	Black	Brown	Buff	Chocolate	Green	Orange	Purple	White	Yellow	**Total**
0	1,648	1,744	48	48	–	48	48	576	48	**4,208**
	39.16%	41.44%	1.14%	1.14%	–	1.14%	1.14%	13.69%	1.14%	
1	224	224	–	1,584	72	–	–	1,812	–	**3,916**
	5.72%	5.72%	–	40.45%	1.84%	–	–	46.27%	–	
Total	1,872	1,968	48	1,632	72	48	48	2,388	48	**8,124**

Table 4: Cross Tabulation of Class by Spore Print Color

Class	black	brown	buff	chocolate	gray	green	orange	pink	purple	...	**Total**
0	344	936	–	204	248	–	64	852	444	...	**4,208**
	8.18%	22.24%	–	4.85%	5.89%	–	1.52%	20.25%	10.55%		
1	64	112	1,728	528	504	24	–	640	48	...	**3,916**
	1.63%	2.86%	44.13%	13.48%	12.87%	0.61%	–	16.34%	1.23%		
Total	768	144	1,584	1,728	1,504	576	24	947	1,440	...	**8,124**

Table 5: Cross Tabulation of Class by Gill Color

Class	evanescent	flaring	large	none	pendant
0	1008 23.95%	48 1.14%	– –	– –	3,152 74.91%
1	1,768 45.15%	– –	1,296 33.10%	36 0.92%	816 20.84%
Total	2,776	48	1,296	36	3,968

Table 6: Cross Tabulation of Class by Ring Type

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